

Marketing Intelligence Agent v2

Demo Walkthrough

Assignment 4

Project Overview

The Marketing Intelligence Agent v2 is an enhanced n8n workflow that collects marketing articles from multiple RSS feeds, analyzes them using Google Gemini AI, and generates human-readable CSV reports.

Compared with Assignment 3, this version introduces AI-powered analysis, random article selection, improved error handling, and workflow scale testing.

Key capabilities include:

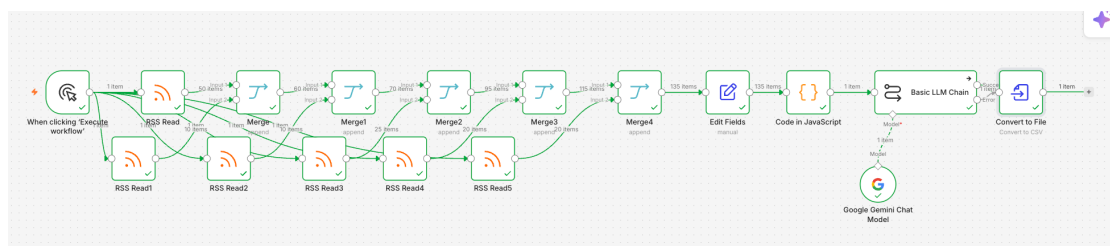
- 6 RSS sources integrated
- Google Gemini AI analysis
- Content category detection
- Priority scoring
- Marketing recommendations
- Error handling for API failures
- Human-readable CSV output generation

Step 1: Execute the Workflow

The workflow starts with a manual trigger and collects articles from multiple RSS feeds.

The system merges all incoming data, standardizes the fields, and prepares the content for AI processing.

Proof



Step 2: AI Analysis

The selected article is sent to Google Gemini AI.

The AI model generates:

- Category
- Priority
- One Sentence Summary
- Marketing Recommendation

This transforms raw RSS data into structured marketing intelligence.

Proof

The screenshot displays the 'Basic LLM Chain' workflow in a software interface. The left sidebar shows a list of nodes: 'Limit' (10 items), 'Edit Fields' (135 items), 'Merge4' (135 items), and 'Merge3' (115 items). The main workspace is divided into 'Parameters' and 'Settings' tabs. The 'Parameters' tab is active, showing a 'Prompt (User Message)' field with the text: 'You are a marketing intelligence assistant. Analyze the following article.' Below this, there are checkboxes for 'Require Specific Output Format' and 'Enable Feedback Model'. The 'Output' tab on the right shows an error message: 'The service is receiving too many requests from you'. The error details include a GoogleGenerativeAI error message and a JSON object with the following structure:

```
{
  "type": "type.googleapis.com/google.rpc.Help",
  "links": [
    {
      "description": "Learn more about Gemini API quotas",
      "url": "https://ai.google.dev/gemini-api/docs/rate-limits"
    }
  ],
  "quotaMetric": "generativelanguage.googleapis.com/generate_content_free_tier_requests",
  "quotaDimensions": {
    "location": "global",
    "model": "gemini-2.5-flash-lite",
    "quotaValue": "20"
  },
  "type": "type.googleapis.com/google.rpc.RetryInfo",
  "retryDelay": "57s"
}
```

Step 3: Generate Human-Readable Output

After AI processing, the workflow exports the result into a CSV file.

The generated file can be opened by non-technical users and used directly for marketing research.

Example output includes:

- Content Marketing
- SEO Analysis
- AI Video Editing
- Chatbot Building
- Deep Research

Proof

Category: Content Marketing

Priority: High

One Sentence Summary: New research from HubSpot indicates that search engines favor specific on-page content formats that are more likely to earn citations and drive organic traffic.

Marketing Recommendation: Prioritize creating and optimizing content in formats such as lists, how-to guides, and articles that offer in-depth analysis and actionable advice, as these are demonstrably favored by search engines for earning citations and ranking well.

Step 4: Error Handling

During testing, Google Gemini occasionally returned:

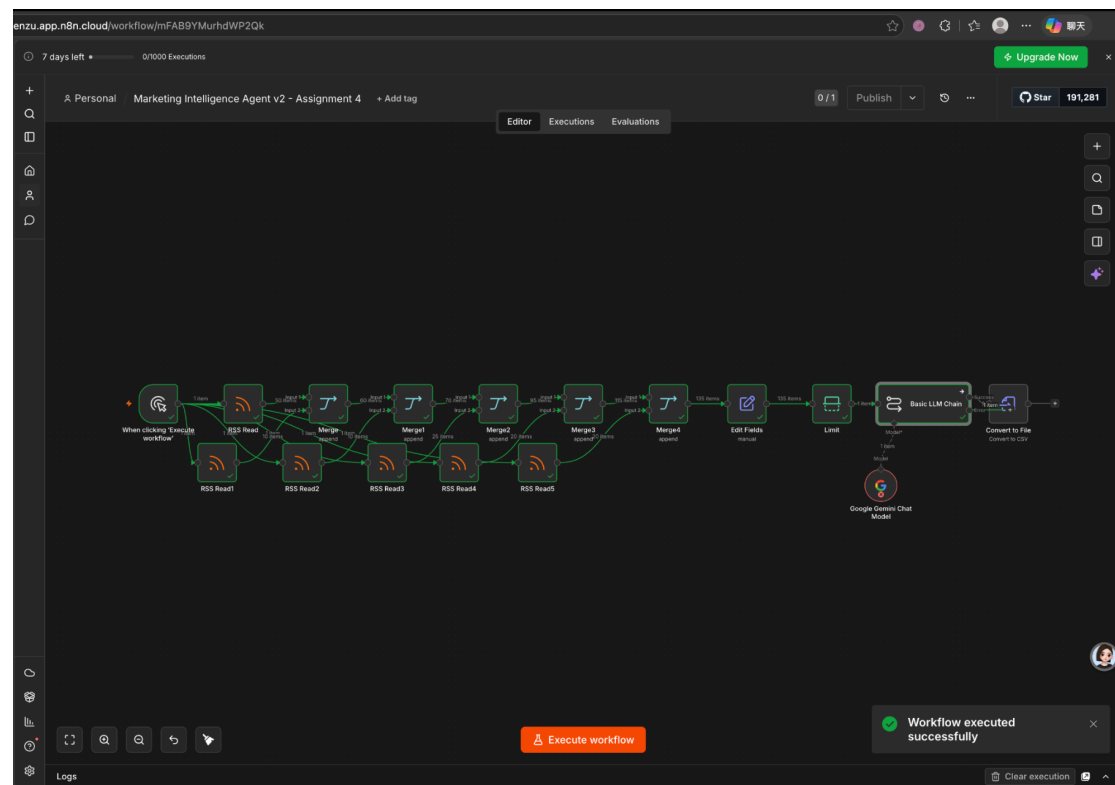
429 Too Many Requests

Instead of stopping the workflow, the error branch captured the failure and continued processing.

An error CSV file was generated automatically.

This improvement increased workflow reliability and demonstrated real-world resilience.

Proof



Step 5: Output Gallery

Multiple workflow executions produced a collection of final usable outputs.

Generated files include:

- 13 successful marketing analysis CSV reports
- 3 API error handling CSV reports

All outputs were downloaded and verified manually.

Proof

outputs/			
名称			
	marketing_analysis_error_handling_example1.csv		
	marketing_analysis_error_handling_example2.csv		
	marketing_analysis_error_handling_example3.csv		
	marketing_analysis_success_example1.csv		
	marketing_analysis_success_example2.csv		
	marketing_analysis_success_example3.csv		
	marketing_analysis_success_example4.csv		
	marketing_analysis_success_example5.csv		
	marketing_analysis_success_example6.csv		
	marketing_analysis_success_example7.csv		
	marketing_analysis_success_example8.csv		
	marketing_analysis_success_example9.csv		
	marketing_analysis_success_example10.csv		
	marketing_analysis_success_example11.csv		
	marketing_analysis_success_example12.csv		
	marketing_analysis_success_example13.csv		

Scale Testing Results

The workflow was tested under multiple real-world scenarios.

Successful Tests

- Multiple workflow executions
- AI analysis completed
- CSV export completed
- Human-readable reports generated

Error Tests

- API rate limit reached

- Error captured successfully
- Workflow continued running
- Error CSV generated

Lessons Learned

This project demonstrated that external AI services may fail unexpectedly because of API limits and quota restrictions.

Instead of stopping completely, the workflow was redesigned to preserve useful information and continue processing.

The final system is more reliable, scalable, and closer to a real-world automation pipeline.

Final Outcome

The enhanced Marketing Intelligence Agent v2 successfully demonstrates:

- ✓ AI-enhanced workflow
- ✓ Multi-source RSS aggregation
- ✓ Error handling and recovery
- ✓ Scale testing
- ✓ Human-readable CSV outputs
- ✓ Real-world workflow resilience